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Methods

Datasets and candidate generation

Snapshots of available datasets. GNoME discoveries aim to extend the catalogues of known stable crystals. In particular, we build off previous work by the Materials Project¹⁶, the OQMD¹⁷, Wang, Botti and Marques (WBM)²⁷ and the ICSD¹⁵. For reproducibility, GNoME-based discoveries use snapshots of the two datasets saved at a fixed point in time. We use the data from the Materials Project as of March 2021 and the OQMD as of June 2021. These structures are used as the basis for all discovery including via SAPS, yielding the catalogue of stable crystals as a result of GNoME. Further updates and incorporation of discoveries by these two groups could yield an even greater number of crystal discoveries.

For a revised comparison, another snapshot of the Materials Project, the OQMD and WBM was taken in July 2023. Approximately 216,000 DFT calculations were performed at consistent settings and used to compare the rate of GNoME discoveries versus the rate of discoveries by concurrent research efforts. From 2021 to 2023, the number of stable crystals external to GNoME expanded from 35,000 to 48,000, relatively small in comparison with the 381,000 new stable crystal structures available on the convex hull presented in this paper.

Substitution patterns. Structural substitution patterns are based on data-mined probabilities from ref. 22. That work introduced a probabilistic model for assessing the likelihood for ionic species substitution within a single crystal structure. In particular, the probability of substitution is calculated as a binary feature model such that $p(X, X') \approx \frac{\exp \sum_i \lambda_i f_i^{(n)}(X, X')}{Z}$, in which X and X' are n -component vectors of n different ions. The model is simplified so that f_i is 0 or 1 if a specific substitution pair occurs and λ_i provides a weighting for the likelihood of a given substitution. The resulting probabilities have been helpful, for example, in discovering new quaternary ionic compounds with limited computation budgets.

In our work, we adjust the probabilistic model so as to increase the number of candidates and give priority to discovery. In particular, the conditional probability computation in the original substitution patterns prefers examples that are more likely to be found in the original dataset. For example, any uncommon element is assigned a smaller probability in the original model. To give priority to novel discovery and move further away from the known sets of stable crystals, we modify the implementation so that probabilities are only computed when two compositions differ. This minor modification has substantial benefits across our pipeline, especially when scaling up to six unique elements.

We also introduce changes to the model parameters to promote novel discovery. In the original probabilistic model, positive lambda refers to more likely substitutions, although ‘unseen’ or uncommon substitution resulted in negative lambda values. We increase the number of generations by setting the minimum value of any substitution pair to be 0. We then threshold high-probability substitutions to a value of 0.001, enabling efficient exploration in composition space through branch-and-bound algorithms available from pymatgen. Overall, these settings allow for many one-ion or two-ion substitutions to be considered by the graph networks that otherwise would not have been considered. We find this to be a good intermediate between the original model and using all possible ionic substitutions, in which we encounter combinatorial blow-ups in the number of candidates.

For the main part of this paper, substitutions are only allowed into compositions that do not match any available compositions in the Materials Project or in the OQMD, rather than comparing structures using heuristic structure matchers. This ensures that we introduce novel compositions in the dataset instead of similar structures that may be missed by structure matchers.

SAPS. To further increase the diversity of structures generations, we introduce a framework that we refer to as symmetry aware partial

substitutions (SAPS), which generalizes common substitution frameworks. For a motivating example, consider the cases of (double) perovskites. Ionic substitutions on crystals of composition $A_2B_2X_6$ does not lead to discovering double perovskites $A_2BB'O_6$, although the two only differ by a partial replacement on the B site.

SAPS enable efficient discovery of such structures. Starting with an original composition, we obtain candidate ion replacements using the probabilities as defined in the ‘Substitution patterns’ section. We then obtain Wyckoff positions of the input structures by means of symmetry analysers available through pymatgen. We enable partial replacements from 1 to all atoms of the candidate ion, for which at each level we only consider unique symmetry groupings to control the combinatorial growth. Early experiments limited the partial substitutions to materials that would charge-balance after partial substitutions when considering common oxidation states; however, greater expansion of candidates was achieved by removing such charge-balancing from the later experiments. This partial-substitution framework enables greater use of common crystal structures while allowing for the discovery of new prototypical structures, as discussed in the main part of this paper. Candidates from SAPS are from a different distribution to the candidates from full substitutions, which increases the diversity of our discoveries and our dataset.

To validate the impact of the SAPS, we traced reference structures from substitutions of all 381,000 novel stable structures back to a structure in the Materials Project or the OQMD by means of a topological sort (necessary as discovered materials were recycled for candidate generation). A total of 232,477 out of the 381,000 stable structures can be attributed to a SAPS substitution, suggesting notable benefit from this diverse candidate-generation procedure.

Oxidation-state relaxations. For the compositional pipeline, inputs for evaluation by machine-learning models must be unique stoichiometric ratios between elements. Enumerating the combinatorial number of reduced formulas was found to be too inefficient, but common strategies to reduce such as oxidation-state balancing was also too restrictive, for example, not allowing for the discovery of $Li_{15}Si_4$. In this paper, we introduce a relaxed constraint on oxidation-state balancing. We start with the common oxidation states from the Semi-conducting Materials by Analogy and Chemical Theory (SMACT)⁵⁷, with the inclusion of 0 for metallic forms. We allow for up to two elements to exist between two ordered oxidation states. Although this is a heuristic approach, it substantially improves the flexibility of composition generation around oxidation-state-balanced ratios.

AIRSS structure generation. Random structures are generated through AIRSS when needed for composition models²⁶. Random structures are initialized as ‘sensible’ structures (obeying certain symmetry requirements) to a target volume and then relaxed through soft-sphere potentials. A substantial number of initializations and relaxations are needed to discover new materials, as different initial structures lead to different minima on the structure–energy landscape. For this paper, we always generate 100 AIRSS structures for every composition that is otherwise predicted to be within 50 meV of stable through composition-only model prediction.

As we describe in Supplementary Note 5, not all DFT relaxations converge for the 100 initializations per composition. In fact, for certain compositions, only a few initializations converge. One of the main difficulties arises from not knowing a good initial volume guess for the composition. We try a range of initial volumes ranging from 0.4 to 1.2 times a volume estimated by considering relevant atomic radii, finding that the DFT relaxation fails or does not converge for the whole range for each composition. Prospective analysis was not able to uncover why most AIRSS initializations fail for certain compositions, and future work is needed in this direction.

Model training and evaluation

Graph networks. For structural models, edges are drawn in the graph when two atoms are closer than an interatomic distance cutoff (4.0 Å for structural models, 5.0 Å for interatomic potentials). Compositional models default to forming edges between all pairs of nodes in the graph. The models update latent node features through stages of message passing, in which neighbour information is collected through normalized sums over edges and representations are updated through shallow MLPs³⁶. After several steps of message passing, a linear readout layer is applied to the global state to compute a prediction of the energy.

Training structural and composition models. Following Roost (representation learning from stoichiometry)⁵⁸, we find GNNs to be effective at predicting the formation energy of a composition and structure.

For the structural models, the input is a crystal definition, which encodes the lattice, structure and atom definitions. Each atom is represented as a single node in the graph. Edges are defined when the interatomic distance is less than a user-defined threshold. Nodes are embedded by atom type, edges are embedded on the basis of the interatomic distance. We also include a global feature that is connected in the graph representation to all nodes. At every step of the GNN, neighbouring nodes and edge features are aggregated and used to update the corresponding representations of nodes, edges or globals individually. After 3–6 layers of message passing, an output layer projects the global vector to get an estimate of the energy. All data for training are shifted and scaled to approximately standardize the datasets. This structural model trained on the Materials Project data obtains state-of-the-art results of a mean absolute error of 21 meV atom⁻¹. Training during the active-learning procedure leads to a model with a final mean absolute error of 11 meV atom⁻¹. Training for structural models is performed with 1,000 epochs, with a learning rate of 5.55×10^{-4} and a linear decay learning rate schedule. By default, we train with a batch size of 256 and use swish nonlinearities in the MLP. To embed the edges, we use a Gaussian featurizer. The embedding dimension for all nodes and edges is 256 and, unless otherwise stated, the number of message-passing iterations is 3.

For the compositional models, the input composition to the GNN is encoded as a set of nodes, for which each element type in the composition is represented by a node. The ratio of the specific element is multiplied with the one-hot vector. For example, SiO₂ would be represented with two nodes, in which one node feature is a vector of zeros and a 1/3 on the 14th row to represent silicon and the other node is a vector of zeros with a 2/3 on the 8th row to represent oxygen. Although this simplified GNN architecture is able to achieve state-of-the-art generalization on the Materials Project (MAE of 60 meV atom⁻¹ (ref. 25)), it does not offer useful predictions for materials discovery, which was also observed by Bartel et al.²⁵. One of the issues with compositional models is that they assume that the training label refers to the ground-state phase of a composition, which is not guaranteed for any dataset. Thus, the formation-energy labels in the training and test sets are inherently noisy, and reducing the test error does not necessarily imply that one is learning a better formation-energy predictor. To explore this, we created our own training set of compositional energies, by running AIRSS simulations on novel compositions. As described in Supplementary Note 5, we find that compositions for which there are only a few completed AIRSS runs tend to have large formation energies, often larger than predicted by the compositional GNN. We find that, if we limit ourselves to compositions for which at least ten AIRSS runs are completed, then the compositional GNN error is reduced to 40 meV atom⁻¹. We then use the GNN trained on such a dataset (for which labels come from the minimum formation energy phase for compositions with at least ten completed AIRSS runs and ignoring the Materials Project data) and are able to increase the precision of stable prediction to 33%.

Model-based evaluation. Discovering new datasets aided by neural networks requires a careful balance between ensuring that the neural networks trained on the dataset are stable and promoting new discoveries. New structures and prototypes will be inherently out of distribution for models; however, we hope that the models are still capable of extrapolating and yielding reasonable predictions. This is out-of-distribution detection problem is further exacerbated by the implicit domain shift, in which models are trained on relaxed structures but evaluated on substitutions before relaxation. To counteract these effects, we make several adjustments to stabilize test-time predictions.

Test-time augmentations. Augmentations at test time are a common strategy for correcting instabilities in machine-learning predictions. Specific to structural models, we especially consider isotropic scaling of the lattice vectors, which both shrinks and stretches bonds. At 20 values ranging from 80% to 120% of the reference lattice scaling volume, we aggregate by means of minimum reduction. This has the added benefit of potentially correcting for predicting on nonrelaxed structures, as isotropic scaling may yield a more appropriate final structure.

Deep ensembles and uncertainty quantification. Although neural network models offer flexibility that allows them to achieve state-of-the-art performance on a wide range of problems, they may not generalize to data outside the training distribution. Using an ensemble of models is a simple, popular choice for providing predictive uncertainty and improving generalization of machine-learning predictions³³. This technique simply requires training n models rather than one. The prediction corresponds to the mean over the outputs of all n models; the uncertainty can be measured by the spread of the n outputs. In our application of training machine-learning models for stability prediction, we use $n = 10$ graph networks. Moreover, owing to the instability of graph-network predictions, we find the median to be a more reliable predictor of performance and use the interquartile range to bound uncertainty.

Model-based filtration. We use test-time augmentation and deep-ensemble approaches discussed above to filter candidate materials based on energy. Materials are then compared with the available GNoME database to estimate the decomposition energy. Note that the structures provided for model-based filtration are unlikely to be completely related, so a threshold of 50 meV atom⁻¹ was used for active learning to improve the recall of stable crystal discovery.

Clustered-based reduction. For active-learning setups, only the structure predicted to have the minimum energy within a composition is used for DFT verification. However, for an in-depth evaluation of a specific composition family of interest, we design clustering-based reduction strategies. In particular, we take the top 100 structures for any given composition and perform pairwise comparisons with pymatgen's built-in structure matcher. We cluster the connected components on the graph of pairwise similarities and take the minimum energy structure as the cluster representation. This provides a scalable strategy to discovering polymorphs when applicable.

Active learning. Active learning was performed in stages of generation and later evaluation of filtered materials through DFT. In the first stage, materials from the snapshots of the Materials Project and the OQMD are used to generate candidates with an initial model trained on the Materials Project data, with a mean absolute error of 21 meV atom⁻¹ in formation energy. Filtration and subsequent evaluation with DFT led to discovery rates between 3% and 10%, depending on the threshold used for discovery. After each round of active learning, new structural GNNs are trained to improve the predictive performance. Furthermore, stable crystal structures are added to the set of materials that can be substituted into, yielding a greater number of candidates to be filtered

by the improved models. This procedure of retraining and evaluation was completed six times, yielding the total of 381,000 stable crystal discoveries. Continued exploration with active learning may continue to drive the number of stable crystals higher.

Composition-based hashing. Previous efforts to learn machine-learning models of energies often use a random split over different crystal structures to create the test set on which energy predictions are evaluated. However, as the GNoME dataset contains several crystal structures with the same composition, this metric is less trustworthy over GNoME. Having several structures within the same composition in both the training and the test sets markedly reduces test error, although the test error does not provide a measure of how well the model generalizes to new compositions. In this paper, we use a deterministic hash for the reduced formula of each composition and assign examples to the training (85%) and test (15%) sets. This ensures that there are no overlapping compositions in the training and test sets. We take a standard MD5 hash of the reduced formula, convert the hexadecimal output to an integer and take modulo 100 and threshold at 85.

DFT evaluation

VASP calculations. We use the VASP (refs. 34,59) with the PBE⁴¹ functional and PAW^{40,60} potentials in all DFT calculations. Our DFT settings are consistent with the Materials Project workflows as encoded in pymatgen²³ and atomate⁶¹. We use consistent settings with the Materials Project workflow, including the Hubbard U parameter applied to a subset of transition metals in DFT+ U , 520 eV plane-wave-basis cutoff, magnetization settings and the choice of PBE pseudopotentials, except for Li, Na, Mg, Ge and Ga. For Li, Na, Mg, Ge and Ga, we use more recent versions of the respective potentials with the same number of valence electrons. For all structures, we use the standard protocol of two-stage relaxation of all geometric degrees of freedom, followed by a final static calculation, along with the custodian package²³ to handle any VASP-related errors that arise and adjust appropriate simulations. For the choice of KPOINTS, we also force gamma-centred kpoint generation for hexagonal cells rather than the more traditional Monkhorst–Pack. We assume ferromagnetic spin initialization with finite magnetic moments, as preliminary attempts to incorporate different spin orderings showed computational costs that were prohibitive to sustain at the scale presented. In AIMD simulations, we turn off spin polarization and use the NVT ensemble with a 2-fs time step.

Bandgap calculations. For validation purposes (such as the filtration of Li-ion conductors), bandgaps are calculated for most of the stable materials discovered. We automate bandgap jobs in our computation pipelines by first copying all outputs from static calculations and using the pymatgen-based MPNonSCFSet in line mode to compute the bandgap and density of states of all materials. A full analysis of patterns in bandgaps of the novel discoveries is a promising avenue for future work.

r²SCAN. r²SCAN is an accurate and numerically efficient functional that has seen increasing adoption from the community for increasing the fidelity of computational DFT calculations. This functional is provided in the upgraded version of VASP6 and, for all corresponding calculations, we use the settings as detailed by MPScanRelaxSet and MPScanStaticSet in pymatgen. Notably, r²SCAN functionals require the use of PBE52 or PBE54 potentials, which can differ slightly from the PBE equivalents used elsewhere in this paper. To speed up computation, we perform three jobs for every SCAN-based computation. First, we precondition by means of the updated PBE54 potentials by running a standard relaxation job under MPRelaxSet settings. This preconditioning step greatly speeds up SCAN computations, which—on average—are five times slower and can otherwise crash on our infrastructure owing to elongated trajectories. Then, we relax with the r²SCAN functional, followed by a static computation.

Metrics and analysis methodology

Decomposition energies. To compute decomposition energies and count the total number of stable crystals relative to previous work^{16,17} in a consistent fashion, we recalculated energies of all stable materials in the Materials Project and the OQMD with identical, updated DFT settings as enabled by pymatgen. Furthermore, to ensure fair comparison and that our discoveries are not affected by optimization failures in these high-throughput recalculations, we use the minimum energy of the Materials Project calculation and our recalculation when both are available.

Prototype analysis. We validate the novel discoveries using XtalFinder (ref. 39), using the compare_structures function available from the command line. This process was parallelized over 96 cores for improved performance. We also note that the symmetry calculations in the built-in library fail on less than ten of the stable materials discovered. We disable these filters but note that the low number of failures suggests minimal impact on the number of stable prototypes.

Families of interest. Layered materials. To count the number of layered materials, we use the methodology developed in ref. 45, which is made available through the pymatgen.analysis.dimensionality package with a default tolerance of 0.45 Å.

Li-ion conductors. The estimated number of viable Li-ion conductors reported in the main part of this paper is derived using the methodology in ref. 46 in a high-throughput fashion. This methodology involves applying filters based on bandgaps and stabilities against the cathode Li-metal anode to identify the most viable Li-ion conductors.

Li/Mn transition-metal oxide family. The Li/Mn transition-metal oxide family is discussed in ref. 25 to analyse the capabilities of machine-learning models for use in discovery. In the main text, we compare against the findings in the cited work suggesting limited discovery within this family through previous machine-learning methods.

Definition of experimental match. In the main part of this paper, we refer to experimentally validated crystal structures with the ICSD. More specifically, we queried the ICSD in January 2023 after many of crystal discoveries had been completed. We then extracted relevant journal (year) and chemical (structure) information from the provided files. By rounding to nearest integer formulas, we found 4,235 composition matches with materials discovered by GNoME. Of these, 4,180 are successfully parsed for structure. Then, we turn to the structural information provided by the ICSD. We used the CIF parser module of pymatgen to load the experimental ICSD structures into pymatgen and then compared those to the GNoME dataset using its structure matcher module. For both modules, we tried using the default settings as well as more tolerant settings that improve structure parsing and matching (higher occupancy tolerance in CIF parsing to fix cases with >1.0 total occupancy and allowing supercell and subset comparison in matching). The latter resulted in a slight increase (about 100) in the number of matched structures with respect to the default settings. Given that we are enforcing a strict compositional match, our matching process is still relatively conservative and is likely to yield a lower bound. Overall, we found 736 matches, providing experimental confirmation for the GNoME structures. 184 of these structures correspond to novel discoveries since the start of the project.

Methods for creating figures of GNoME model scaling

Figures 1e and 3a,b show how the generalization abilities of GNoME models scale with training set size. In Fig. 1e, the training sets are sampled uniformly from the materials from the Materials Project and from our structural pipeline, which only includes elemental and partial substitutions into stable materials in the Materials Project and the OQMD. The training labels are the final formation energy at the end of relaxation. The test set is constructed by running AIRSS on 10,000 random

compositions filtered by the SMACT. Test labels are the final formation energy at the end of the AIRSS relaxation, for crystals that AIRSS and DFT (both electronically and ionically) converged. Because we apply the same composition-based hash filtering (see ‘Composition-based hashing’ section) on all of our datasets, there is no risk of label leakage between the training set from the structural pipeline and the test set from AIRSS.

In Fig. 3a, we present the classification error for predicting the outcome of DFT-based molecular dynamics using GNN molecular dynamics. ‘GNoME: unique structures’ refers to the first step in the relaxation of crystals in the structural pipeline. We train on the forces on each atom on the first DFT step of relaxation. The different training subsets are created by randomly sampling compositions in the structural pipeline uniformly. ‘GNoME: intermediate structures’ includes all the same compositions as ‘GNoME: unique structures’, but has all steps of DFT relaxation instead of just the first step. The red diamond refers to the same GNN interatomic potential trained on the data from M3GNet, which includes three relaxation steps per composition (first, middle and last), as described in the M3GNet paper⁶².

Coding frameworks

For efforts in machine learning, GNoME models make use of JAX and the capabilities to just-in-time compile programs onto devices such as graphics processing units (GPUs) and tensor processing units (TPUs). Graph networks implementations are based on the framework developed in Jraph, which makes use of a fundamental GraphsTuple object (encoding nodes and edges, along with sender and receiver information for message-passing steps). We also make great use of functionality written in JAX MD for processing crystal structures⁶³, as well as TensorFlow for parallelized data input⁶⁴.

Large-scale generation, evaluation and summarization pipelines make use of Apache Beam to distribute processing across a large number of workers and scale to the sizes as described in the main part of this paper (see ‘Overview of generation and filtration’ section). For example, billions of proposal structures, even efficiently encoded, requires terabytes of storage that would otherwise fail on single nodes.

Also, crystal visualizations are created using tooling from VESTA (ref. 65).

MLIPs

Pretrained GNoME potential. We train a NequIP potential³⁰, implemented in JAX using the e3nn-jax library⁶⁶, with five layers, hidden features of $128\ell = 0$ scalars, $64\ell = 1$ vectors and $32\ell = 2$ tensors (all even irreducible representations only, $128x0e + 64x1x + 32x2e$), as well as an edge-irreducible representation of $0e + 1e + 2e$. We use a radial cutoff of 5 Å and embed interatomic distances r_{ij} in a basis of eight Bessel functions, which is multiplied by the XPLOFF cutoff function, as defined in HOOMD-blue (ref. 67), using an inner cutoff of 4.5 Å. We use a radial MLP $R(r)$ with two hidden layers with 64 neurons and a SiLU nonlinearity. We also use SiLU for the gated, equivariant nonlinearities⁶⁸. We embed the chemical species using a 94-element one-hot encoding and use a self-connection, as proposed in ref. 30. For internal normalization, we divide by 26 after each convolution. Models are trained with the Adam optimizer using a learning rate of 2×10^{-3} and a batch size of 32. Given that high-energy structures in the beginning of the trajectory are expected to be more diverse than later, low-energy structures, which are similar to one another and often come with small forces, each batch is made up of 16 structures sampled from the full set of all frames across all relaxations and 16 structures sampled from only the first step of the relaxation only. We found this oversampling of first-step structures to substantially improve performance on downstream tasks. The learning rate was decreased to a new value of 2×10^{-4} after approximately 23 million steps, to 5×10^{-5} after a further approximately 11 million steps and then trained for a final 2.43 million steps. Training was performed on four TPU v3 chips.

We train on formation energies instead of total energies. Formation energies and forces are not normalized for training but instead we predict the energy as a sum over scaled and shifted atomic energies, such that $\hat{E} = \sum_{i \in N_{\text{atoms}}} (\hat{e}_i \sigma + \mu)$, in which $\hat{e}_i$ is the final, scalar node feature on atom i and σ and μ are the standard deviation and mean of the per-atom energy computed over a single pass of the full dataset. The network was trained on a joint loss function consisting of a weighted sum of a Huber loss on energies and forces:

$$\mathcal{L} = \lambda_E \frac{1}{N_b} \sum_{b=1}^{b=N_b} \mathcal{L}_{\text{Huber}} \left(\delta_E, \frac{\hat{E}_b}{N_a}, \frac{E_b}{N_a} \right) + \lambda_F \frac{1}{N_b} \sum_{b=1}^{b=N_b} \sum_{a=1}^{a=N_a} \mathcal{L}_{\text{Huber}} \left(\delta_F, -\frac{\partial \hat{E}_b}{\partial r_{b,a,\alpha}}, F_{b,a,\alpha} \right) \quad (1)$$

in which N_a and N_b denote the number of atoms in a structure and the number of samples in a batch, respectively, $\hat{E}_b$ and E_b are the predicted and true energy for a given sample in a batch, respectively, and $F_{a,\alpha}$ is the true force component on atom a , for which $\alpha \in \{x, y, z\}$ is the spatial component. $\mathcal{L}_{\text{Huber}}(\delta, \hat{a}, a)$ denotes a Huber loss on quantity a , for which we use $\delta_E = \delta_F = 0.01$. The pretrained potential has 16.24 million parameters. Inference on an A100 GPU on a 50-atom system takes approximately 14 ms, enabling a throughput of approximately 12 ns day^{-1} at a 2-fs time step, making inference times highly competitive with other implementations of GNN interatomic potentials. Exploring new approaches with even further improved computational efficiency is the focus of future work.

Training on M3GNet data. To allow a fair comparison with the smaller M3GNet dataset used in ref. 62, a NequIP model was trained on the M3GNet dataset. We chose the hyperparameters in a way that balances accuracy and computational efficiency, resulting in a potential with efficient inference. We train in two setups, one splitting the training and testing sets based on unique materials and the other over all structures. In both cases, we found the NequIP potential to perform better than the M3GNet models trained with energies and forces (M3GNet-EF) reported in ref. 62. Given this improved performance, to enable a fair comparison of datasets and dataset sizes, we use the NequIP model trained on the structure-split M3GNet data in the scaling tests (the pretrained M3GNet model is used for zero-shot comparisons). We expect our scaling and zero-shot results to be applicable to a wide variety of modern deep-learning interatomic potentials.

The structural model used for downstream evaluation was trained using the Adam optimizer with a learning rate of 2×10^{-3} and a batch size of 16 for a total of 801 epochs. The learning rate was decreased to 2×10^{-4} after 601 epochs, after which we trained for another 200 epochs. We use the same joint loss function as in the GNoME pretraining, again with $\lambda_E = 1.0$, $\lambda_F = 0.05$ and $\delta_E = \delta_F = 0.01$. The network hyperparameters are identical to the NequIP model used in GNoME pretraining. To enable a comparison with ref. 62, we also subtract a linear compositional fit based on the training energies from the reference energies before training. Training was performed on a set of four V100 GPUs.

AIMD conductivity experiments. Following ref. 69, we classify a material as having superionic behaviour if the conductivity σ at the temperature of 1,000 K, as measured by AIMD, satisfies $\sigma_{1,000\text{K}} > 101.18 \text{ mScm}^{-1}$. Refer to the original paper for applicable calculations. See Supplementary Information for further details.

Robustness experiments. For the materials selected for testing the robustness of our models, $\text{As}_{24}\text{Ca}_{24}\text{Li}_{24}$, $\text{Ba}_8\text{Li}_{16}\text{Se}_{32}\text{Si}_8$, $\text{K}_{24}\text{Li}_{16}\text{P}_{24}\text{Sn}_8$ and $\text{Li}_{32}\text{S}_{24}\text{Si}_4$, a series of models is trained on increasing training set sizes sampled from the $T = 400 \text{ K}$ AIMD trajectory. We then evaluate these models on AIMD data sampled at both $T = 400 \text{ K}$ (to measure the effect of fine-tuning on data from the target distribution) and